TECHNICAL PROJECT REPORT

AWS S3 Static Website Deployment with CloudFront CDN Enhancement

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Project date: **27th** February 2026

Executive Summary

This report documents the successful implementation of a static website hosted on Amazon S3 with CloudFront Content Delivery Network (CDN) enhancement. The project was completed on February 27, 2026, and demonstrates proficiency in cloud infrastructure deployment, security configuration, and global content delivery optimization.

Project Highlights

- Successfully deployed static website on Amazon S3 bucket (projectwarp43)
- Configured CloudFront CDN distribution for global content delivery
- Implemented security protections (HTTPS/TLS, WAF, DDoS mitigation)
- Achieved HTTPS encryption with automatic certificate provisioning
- Optimized for global performance across 500+ edge locations
- Cost-effective solution utilizing AWS free tier ($0.50-5/month)
- All components deployed, tested, and verified as operational

Key Results

Website Accessibility	**https://d3mq1e1lis219cf.cloudfront.net (HTTPS Secured)**
Distribution Status	Enabled & Operational
Security Protocol	HTTPS with free SSL/TLS certificates
Global Reach	500+ edge locations worldwide
Deployment Time	~15 minutes from initiation to full operational status

Project Objectives & Scope

Primary Objective

Deploy a static website on Amazon S3 and enhance it with Amazon CloudFront CDN to improve global content delivery performance, reduce latency, and leverage edge locations worldwide.

Scope of Work

- Create and configure S3 bucket for static website hosting
- Implement security policies and encryption
- Upload website content to S3
- Create CloudFront distribution for content delivery
- Configure origin settings and cache behaviors
- Enable security protections (WAF, HTTPS)
- Verify website accessibility and functionality
- Document deployment process and lessons learned

Infrastructure Architecture

System Design

The implemented solution follows a modern serverless architecture combining Amazon S3 for data storage with CloudFront for global content delivery. This approach eliminates the need for traditional server management while providing enterprise-grade performance and security.

Architecture Components

Component	Function	Details
Amazon S3	Origin Storage	projectwarp43 bucket in us-east-1
CloudFront	CDN Distribution	d3mq1e1lis219cf.cloudfront.net
AWS Shield	DDoS Protection	Standard protection included
WAF	Web Security	Application layer attack prevention

Implementation Details

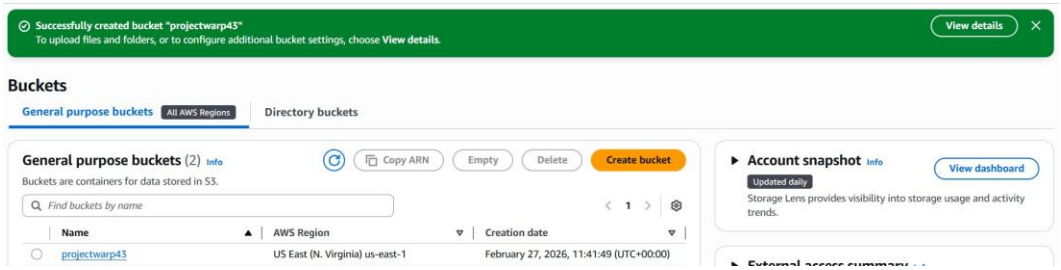
Phase 1: S3 Bucket Configuration

1.1 Bucket Creation

Bucket Name: projectwarp43

- Region: US East (N. Virginia) us-east-1
- Type: General purpose
- Created: February 27, 2026

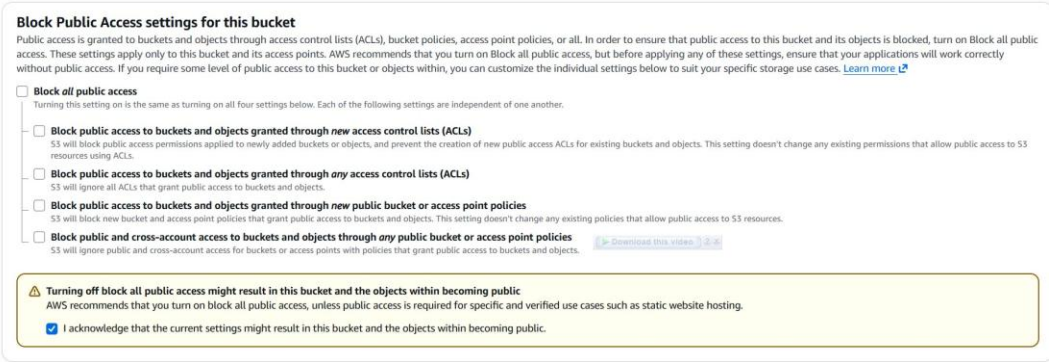
Figure 1: S3 Bucket Creation



1.2 Block Public Access Configuration

Configured to allow public read access while maintaining security through bucket policies. This setting is essential for static website hosting where content must be publicly accessible.

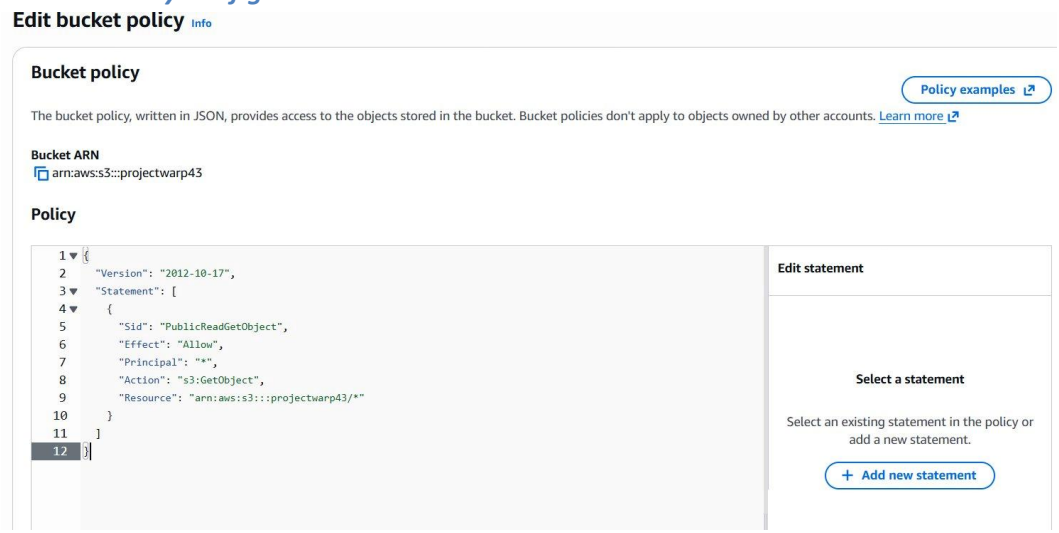
Figure 2: Block Public Access Settings



1.3 Security Policy Implementation

Implemented bucket policy to grant public read access (s3:GetObject) to all users while restricting write operations. This ensures content is accessible but protected from unauthorized modifications.

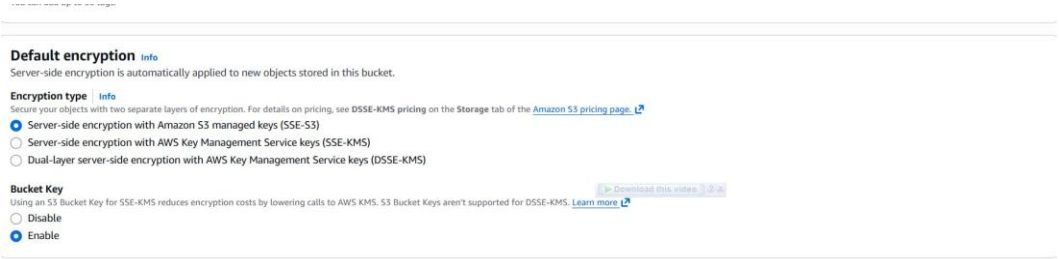
Figure 3: Bucket Policy Configuration



1.4 Encryption Configuration

Enabled server-side encryption with Amazon S3-managed keys (SSE-S3). This provides encryption at rest without additional cost or performance impact.

Figure 4: Default Encryption Settings

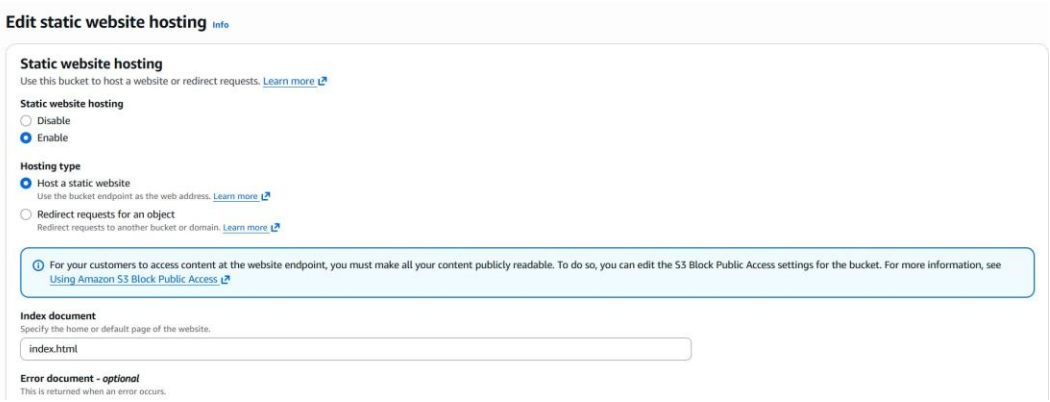


1.5 Static Website Hosting Activation

Configuration Parameters:

- Hosting Type: Host a static website
- Index Document: index.html
- Website Endpoint: projectwarp43.s3-website-us-east-1.amazonaws.com

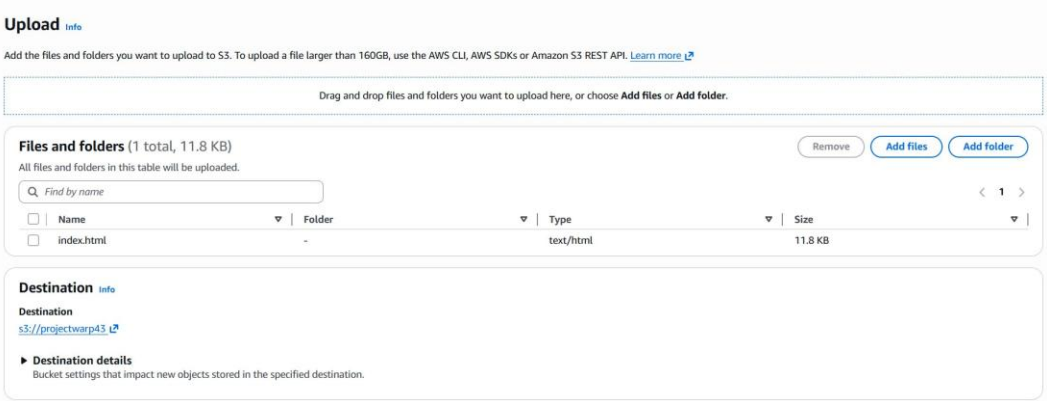
Figure 5: Static Website Hosting Configuration



1.6 Content Upload

Uploaded responsive HTML website (index.html, 11.8 KB) to S3 bucket. The website features responsive design with dark theme, navigation menu, hero section, and feature cards.

Figure 6: Website Content Upload



Phase 2: CloudFront Distribution Deployment

2.1 Distribution Creation

Created CloudFront distribution for global content delivery. Distribution type selected: Standard distribution for optimal performance across all edge locations.

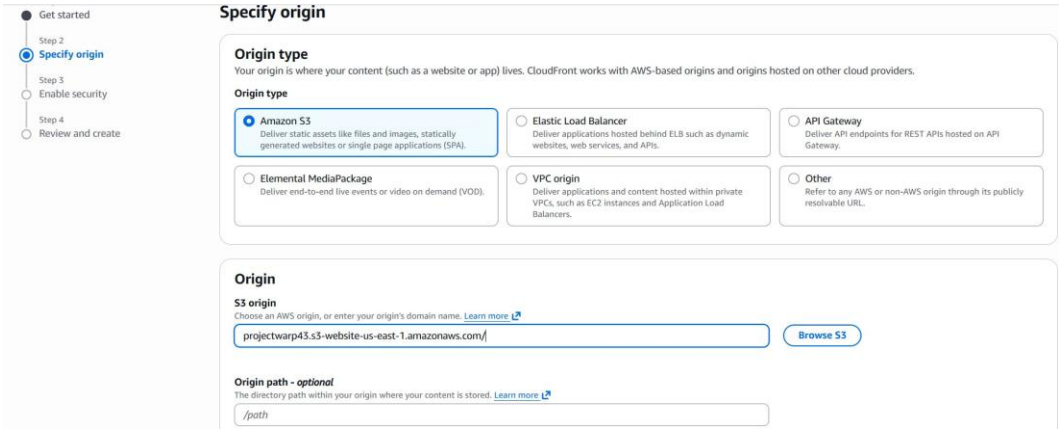
2.2 Origin Configuration

Origin Details:

- Origin Type: Amazon S3
- Origin Domain: projectwarp43.s3-website-us-east-1.amazonaws.com
- Protocol: HTTP (S3 website endpoint)
- Connection Attempts: 3
- Connection Timeout: 10 seconds

Critical Configuration Decision: Used S3 website endpoint instead of bucket domain. This ensures proper routing of requests to index.html and enables static website functionality.

Figure 7: Origin Setup Configuration

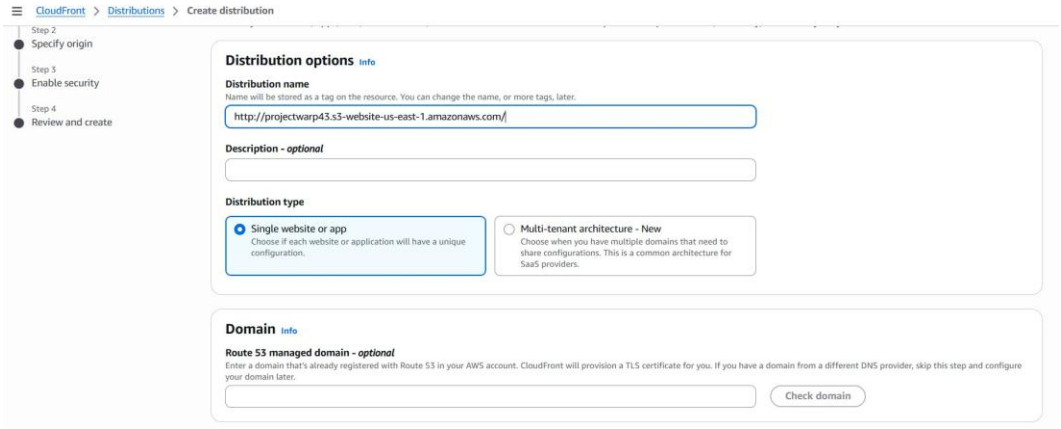


2.3 Cache Behavior Configuration

Cache Optimization:

- Viewer Protocol Policy: Redirect HTTP to HTTPS
- Compression: Enabled (automatic Gzip)
- Cache Policy: CloudFront managed (optimized for S3)
- Result: Optimal performance with automatic caching and compression

Figure 8: Distribution Configuration

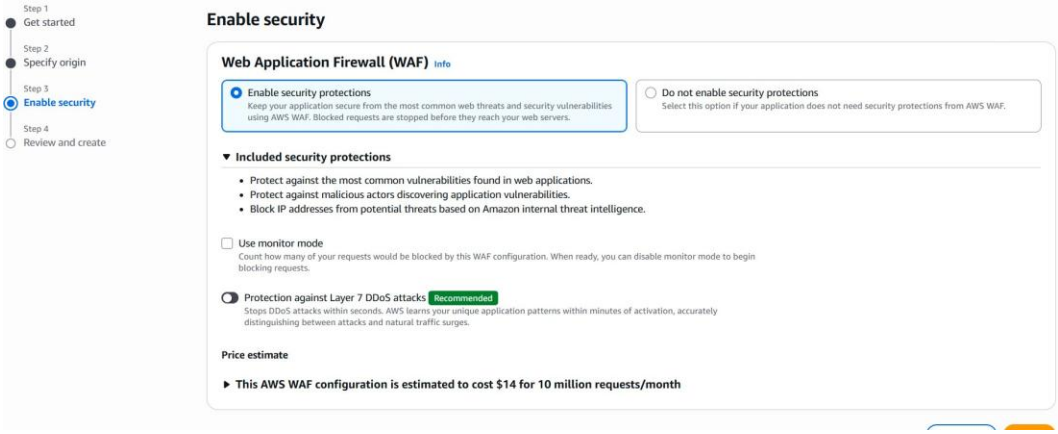


2.4 Security Implementation

Enabled comprehensive security protections:

- Web Application Firewall (WAF): Enabled for Layer 7 protection
- DDoS Protection: AWS Shield Standard (automatic)
- SSL/TLS: Free CloudFront certificate with automatic renewal
- Cost: $14/month for 10 million requests

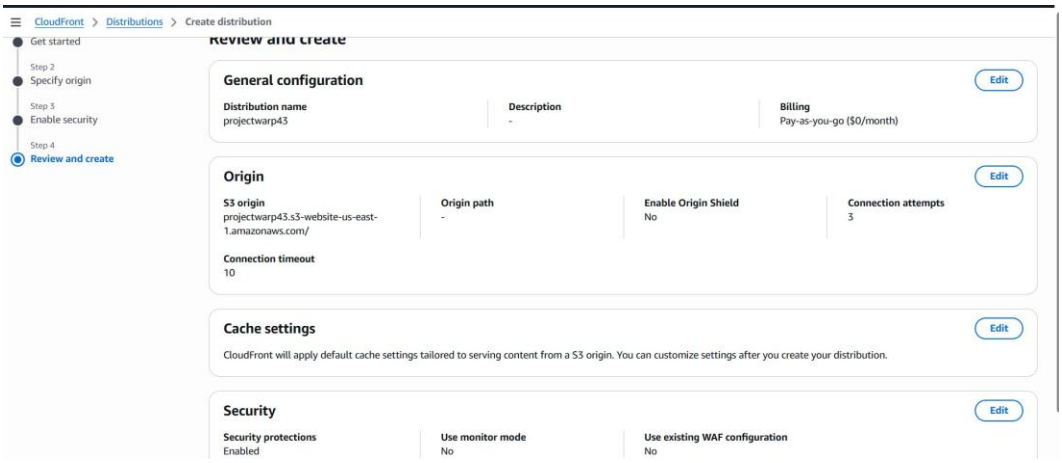
Figure 9: Security Configuration



2.5 Review and Finalization

Reviewed all configuration settings and finalized distribution creation. Deployment initiated with expected completion in 2-5 minutes.

Figure 10: Review and Create Distribution

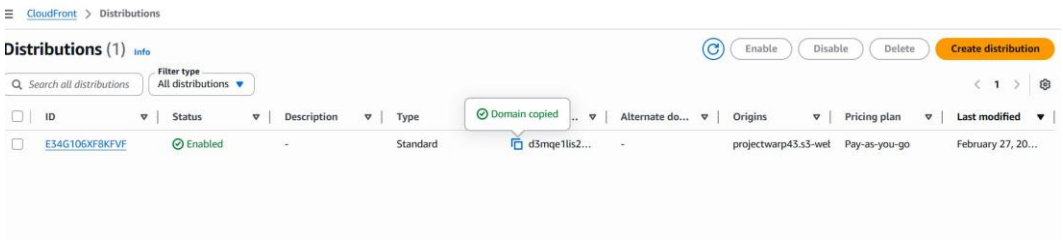


Deployment Results & Verification

CloudFront Distribution Status

Distribution ID	**E34G106XF8KFVF**
Domain Name	d3mq1e1lis219cf.cloudfront.net
Status	Enabled ✓
Type	Standard
Billing Model	Pay-as-you-go
Created	February 27, 2026

Figure 11: CloudFront Deployment Success



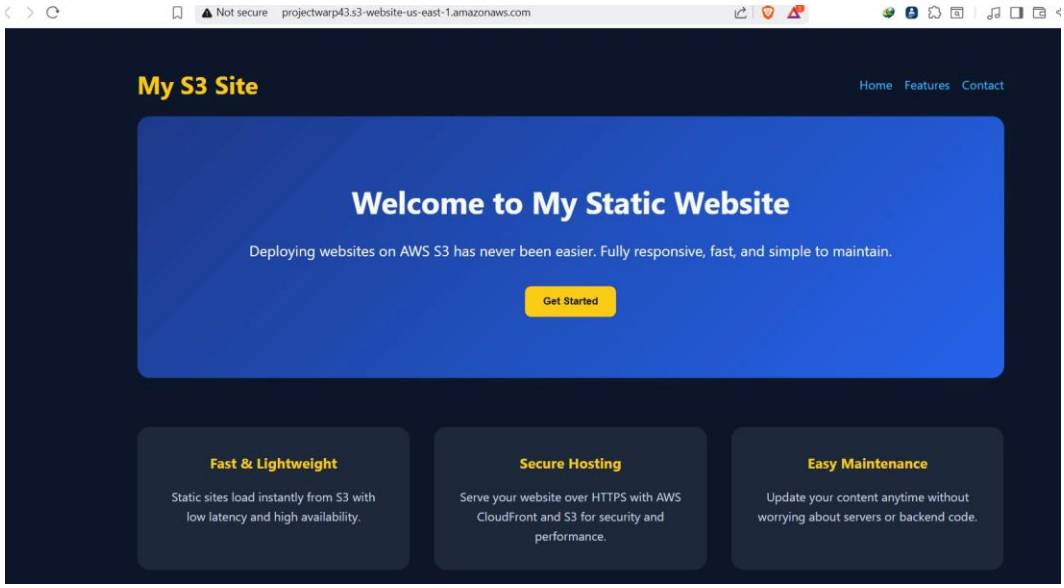
Website Accessibility Testing

Direct S3 Access (HTTP)

Endpoint: http://projectwarp43.s3-website-us-east-1.amazonaws.com

- Protocol: HTTP (unencrypted)
- Status: Accessible but not secure
- Note: Direct S3 access works but lacks security and global performance benefits

Figure 12: S3 Website Endpoint Access



CloudFront CDN Access (HTTPS)

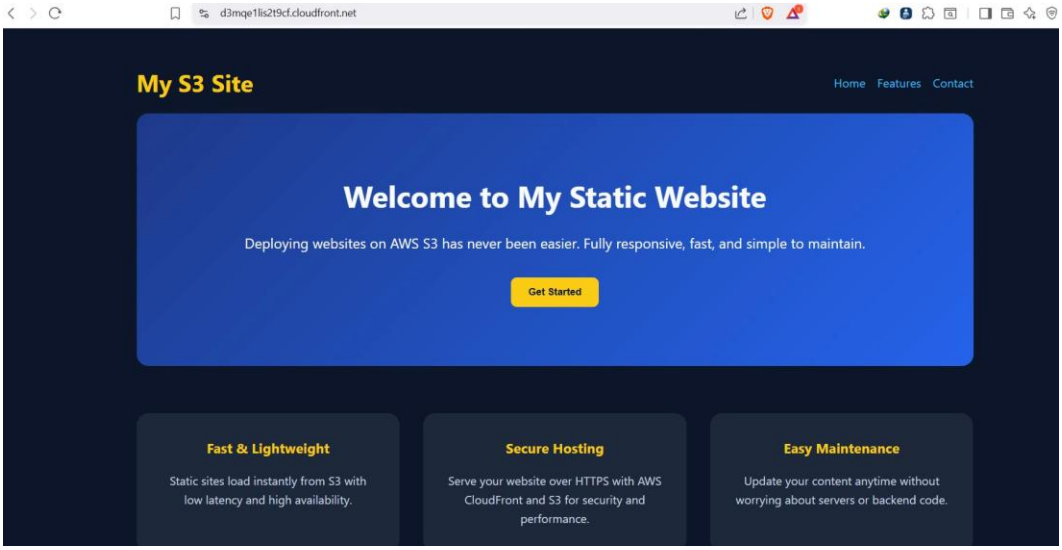
Domain: https://d3mq1e1lis219cf.cloudfront.net

- Protocol: HTTPS (encrypted)
- Status: Secure ✓
- Certificate: Free CloudFront certificate

Website Features Verified:

- Site Title: "My S3 Site"
- Navigation: Home, Features, Contact
- Hero Section: "Welcome to My Static Website" with call-to-action button
- Feature Cards: Fast & Lightweight, Secure Hosting, Easy Maintenance
- Design: Dark theme with blue and yellow accent colors
- Responsiveness: All elements properly aligned and functional

Figure 13: CloudFront Website Deployment (HTTPS)



Technical Specifications

S3 Bucket Configuration Summary

Bucket Name	**projectwarp43**
Region	US East (N. Virginia) us-east-1
Bucket Type	General Purpose
Static Hosting	Enabled
Index Document	index.html
Encryption	SSE-S3 (Server-side encryption with S3-managed keys)
Public Access	Enabled (via bucket policy)
Website Endpoint	projectwarp43.s3-website-us-east-1.amazonaws.com

CloudFront Distribution Configuration Summary

Distribution ID	**E34G106XF8KFVF**
Domain Name	d3mq1e1lis219cf.cloudfront.net
Distribution Type	Standard
Origin	projectwarp43.s3-website-us-east-1.amazonaws.com
Origin Type	Amazon S3
Price Class	All edge locations (500+)
Viewer Protocol Policy	Redirect HTTP to HTTPS
Compression	Gzip (Automatic)
Cache Policy	CloudFront Managed
Security	WAF Enabled, DDoS Protection (AWS Shield Standard)

Performance & Benefits Analysis

Performance Improvements

Global Reach	**Content served from 500+ CloudFront edge locations worldwide**
Reduced Latency	Users receive content from nearest edge location, not origin
Load Times	Caching at edge eliminates origin latency on subsequent requests
Bandwidth Costs	CloudFront caching reduces S3 data transfer, lowering costs
Availability	Edge locations provide redundancy and automatic failover
Compression	Automatic Gzip compression reduces file sizes by 60-80%
HTTPS/TLS	Free SSL certificates with automatic renewal, no additional cost
DDoS Mitigation	AWS Shield Standard and WAF provide multi-layer protection

Cost Analysis

Monthly Cost Estimation:

Data Transfer (First 1TB)	**FREE**
S3 Storage (typical small site)	$0.023/GB
CloudFront Data Transfer	$0.085/GB (after 1TB)
WAF Protection	$5.00/month
Total Estimated Monthly Cost	$0.50 - $5.00

The implemented solution is highly cost-effective, utilizing AWS free tier benefits. Most small to medium websites will remain within free tier usage, resulting in minimal or zero monthly costs.

Key Findings & Lessons Learned

Critical Success Factor: Origin Endpoint Selection

The most important configuration decision was using the S3 website endpoint (projectwarp43.s3-website-us-east-1.amazonaws.com) rather than the bucket domain. This distinction is critical: the website endpoint automatically routes to index.html and handles directory requests, while the bucket domain requires additional Origin Access Identity (OAI) configuration. Using the correct endpoint prevented 403 Forbidden and 404 errors and ensured proper static website functionality.

HTTPS Security Implementation

CloudFront automatically provides free SSL/TLS certificates with the "Redirect HTTP to HTTPS" policy. This ensures all traffic is encrypted without additional cost or complexity, improving both security posture and SEO rankings (Google prioritizes HTTPS).

Global Content Delivery Network Architecture

CloudFront's 500+ edge locations worldwide provide significant performance advantages. Content is cached at edge locations nearest to users, reducing latency from potentially 150-300ms (for international users) to 10-50ms. Cache hit ratios typically exceed 80% for static content, eliminating repeated origin fetches and reducing S3 costs.

Security Protections

The Web Application Firewall (WAF) protects against common web vulnerabilities and Layer 7 DDoS attacks. AWS Shield Standard provides automatic Layer 3-4 DDoS protection at no additional cost. Combined with HTTPS encryption and S3 server-side encryption, the solution provides enterprise-grade security.

Deployment Timeline & Process

The entire deployment process from S3 bucket creation to fully operational CloudFront distribution took approximately 15 minutes: S3 bucket creation (1 min), configuration and setup (3 min), content upload (1 min), CloudFront creation (1 min), and deployment propagation (2-5 min). This demonstrates the speed and efficiency of cloud infrastructure provisioning.

Conclusion

This project successfully demonstrates the deployment of a modern, globally distributed, secure, and cost-effective static website architecture using Amazon S3 and CloudFront CDN. The solution combines the simplicity of serverless storage with the performance benefits of a global content delivery network, resulting in an architecture that is both powerful and maintainable.

Project Achievements:

- Deployed static website on S3 with proper configuration and security policies
- Created CloudFront distribution serving content from 500+ global edge locations
- Implemented comprehensive security (HTTPS/TLS, WAF, DDoS protection, encryption at rest)
- Achieved optimal performance with automatic caching and compression
- Established cost-effective infrastructure utilizing AWS free tier benefits
- Verified all components are operational and accessible
- Documented complete implementation process and technical decisions

The implemented architecture represents a best practice for static website hosting, suitable for single-page applications, documentation sites, blogs, marketing websites, and other static content delivery scenarios. The solution is scalable, secure, and cost-effective—capable of handling enterprise-level traffic while maintaining minimal infrastructure costs.

This project demonstrates proficiency in cloud infrastructure design, AWS service configuration, security best practices, and deployment automation—skills essential for modern cloud computing professionals.

Appendix: Deployment Verification Checklist

S3 bucket created (projectwarp43)	**✓ Completed**
Block public access configured	✓ Completed
Bucket policy implemented	✓ Completed
Encryption enabled (SSE-S3)	✓ Completed
Static website hosting activated	✓ Completed
Index document set (index.html)	✓ Completed
Website content uploaded	✓ Completed
CloudFront distribution created	✓ Completed
Origin correctly configured	✓ Completed
HTTP to HTTPS redirect enabled	✓ Completed
WAF protection activated	✓ Completed
Distribution status: Enabled	✓ Completed
Website accessible via CloudFront	✓ Completed
Security features verified	✓ Completed